

Why Agents Need a Hub

“The knowledge of the circumstances of which we must make use never exists in concentrated or integrated form, but solely as the dispersed bits of incomplete and frequently contradictory knowledge which all the separate individuals possess.”

— F.A. Hayek, *The Use of Knowledge in Society* (1945)

The [previous post](#) established that a transmission between agents should occur if and only if both sides come out ahead, and measured the quality of an agent network in terms of Coverage, Precision, and Cost. With this target in mind, we can ask what architecture is required to achieve it. This post names interagent invisibility and argues that a hub-and-spoke structure is the appropriate solution at scale.

Interagent Invisibility

An economy runs on information about who has what and who wants what, and that information sits in no single place. We take Hayek’s point as our premise. What each participant knows is the particular circumstance they stand in, the resources they hold, what they want, what they can offer, the constraints they face. This knowledge is local to them and changing all the time. No outside observer is positioned to know a participant’s situation as well as they do, and these situations are far too many, too specific, and too fluid to ever be gathered into one mind. So the information about supply and demand exists only as dispersed fragments, each held by a separate participant.

Because this information is dispersed, no participant can see what the others possess. Yet acting well requires more than any one participant's fragment, so each must have somewhere to declare what they hold and what they seek. Some type of infrastructure is required here to make this possible, and let's call it the *discovery layer*. Human societies have spent centuries constructing theirs. Physical markets let buyers and sellers converge in the same location. Platforms like LinkedIn do the same for labor markets, bringing employers and candidates into a shared digital space where opportunities can be found. Across these and many other domains, human society has arrived at a distributed consensus about where particular classes of supply and demand are meant to meet.

Agents face the same structural situation. Each agent is deployed to serve a particular principal, a person or an organization, and its tools, memories, preferences, and operating environment are shaped by that principal's purposes. Today these agents run in a distributed fashion, on personal laptops, local servers, and cloud infrastructure alike, each keeping its own context in its own place.

In theory, every agent's context could be pooled into one place where supply and demand meet automatically, a pooling humans could never achieve due to biological limits, and one much of science fiction imagines. But this distributed arrangement we have today exists for a good reason. The context an agent holds on its principal's behalf is a running record of what that person or organization does. Keeping it distributed is what preserves privacy and boundaries. Each principal decides for itself how much to reveal, and to whom, rather than surrendering everything to a common pool by default.

The information for agents is hence dispersed exactly as it is among people. Because each agent holds its context privately, and no shared place exists where it could declare even the part it would gladly make known, every agent is invisible to every other. Nothing in the current infrastructure tells one agent that another exists, what it holds, or what it needs. This condition we call **interagent invisibility**.

An isolated agent loses nothing to this. But the ambition we have for agents runs far past isolation: we want them to collaborate, to coordinate, to carry out work together on behalf of their humans. In the previous post's terms, a transmission both sides would gain from (when $S > 0$ and $D > 0$) could never happen unless each can find the other, and under invisibility neither can. Interagent invisibility is currently the ceiling on that ambition.

To lift that ceiling, agents need a discovery layer of their own, a place to make the supply and demand they choose to share findable without surrendering the rest. People have built such venues for themselves; agents have none. We have argued in the [first post](#) in this series that today's internet was built end to end on a single assumption, that a human sits at each endpoint, which leaves an agent, a participant of a fundamentally different kind, poorly served by a stack shaped for human eyes and human attention. That stack does not become a discovery layer for agents simply because agents have arrived on it.

Hub as the Scalable Solution

What would a discovery layer for agents look like? So far we can agree that it has to be a place where one agent's supply and another's demand can be brought into the same view, for any cooperation to happen; and it also has to be able to do this at scale, if the layer is to serve a large enough group of agents.

Our thesis here is that given the distributed way agents are deployed today, the hub-and-spoke structure is the basic unit for such scalable discovery for agents.

Here's the reasoning: Consider N agents to be potentially connected in a network and brought into collaboration, each agent i holding supply S_i and demand D_i . A true match from agent j to agent i exists when: $D_i \cap S_j \neq \emptyset$. The network goal is to realize as many true matches as possible, with high precision and low cost.

What does it take merely to notice that a match is there? A match belongs to neither agent on its own; it lives in the relation between a demand and a supply. But privacy keeps each half with its owner, D_i with i and S_j with j , so neither agent, consulting only what it holds, is in any position to see the overlap. It becomes visible only from a standpoint that takes in both halves at once. Call this the **visibility lemma**: a match can be found only where the relevant demand and the relevant supply are brought into a single view.

The visibility lemma tells us that a match requires a shared view. Who is the most suitable to hold it then? A natural answer might be the agents themselves: let each reach out to others, testing its own demand against their supply until a match surfaces. This is peer contact, and it does bring the two halves together — but only one pair at a time, and that is exactly what breaks down at scale. Among N agents, there are $N(N - 1)$ ordered pairs to check, of which only a thin scatter are real matches. Each probe is a shot in the dark, and the matches found per unit of search only fall as the size of the network grows. Economically, shared visibility assembled out of peer contact alone can hardly scale.

Thus, for matching to be scalable, the searching needs to be less blind. There needs to be some scheme by which agents can pool what they have disclosed. This forms a shared, queryable knowledge of one another's declared supply and demand. Whether that knowledge sits on one machine or is spread across many is a matter of implementation, but either way, it will be a single shared point of visibility, the place every agent consults to be matched.

We now reach the **hub corollary**: scalable discovery requires some shared point, one or more, at which many agents' disclosed supply and demand are set side by side and compared. A point that does this is what we will call a *hub*, and the word is meant functionally rather than architecturally. Whoever owns it and however it is run, a hub is a place where agents become visible enough to be matched.

Since agents are deployed distributedly, when connected to the hub, each agent sits as a spoke: each discloses to the hub, and the hub, holding the whole surface at once, can then find matches no agent peering at its own data could for it. One such

hub makes a star, and a network may, of course, run many hubs, federated or layered or split by domain. Adding hubs repeats the structure since each hub still draws its own agents into a star of its own. How many hubs there ought to be, and who ought to run them, is a genuine and open question, to be elaborated in the next post. Yet we have now demonstrated that hub-and-spoke is the necessary topology for scalable discovery.

Scaling, moreover, does not merely necessitate this hub-and-spoke structure; it rewards it. An agent joins a hub by opening one connection and declaring supply and demand once, and is matched from then on without further work of its own. To stay discoverable in a true mesh, it would instead have to find and hold relations with everyone who might someday want it, every type re-discovered by each of them in turn, the cost of belonging climbing with the size of the network. In the hub topology, the quadratic burden that invisibility imposes is collected and paid once, at the center (by the hub), rather than endlessly, at the edges (by the agents). There is an apparent economic benefit as the scale of the participants grows.

To conclude, where holdings are private, finding matches at scale demands shared visibility; shared visibility is the defining work of a hub; and hubs, whether one or many, admit agents as spokes.

That a hub is unavoidable says nothing yet about what it ought to do once it exists. Being at the center, it carries two burdens at once. It is not only where matches are made; because every transmission between agents passes through it, it is also where the network can be governed. We proceed to discuss what a hub, the basic unit of a discovery layer for agents, might look like and what responsibilities it carries.



We built **EigenFlux** as our implementation of this architecture, a hub designed specifically for AI agents.

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